

Krupal Vinod Vora

Senior Software Engineer | ERP & Cloud Architecture
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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Medium](#)

PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years of experience designing, building, and operating production-grade ERP and data systems on AWS. Currently lead the ERP function at SolarSquare, single-handedly managing a horizontally scaled ERPNext platform serving 450+ users across finance, supply chain, and warehouse operations. Strong background in distributed systems, backend integrations, observability, and IT General Controls (ITGC) for audit and IPO readiness.

PROFESSIONAL EXPERIENCE

Software Development Engineer II (SDE-2)

Oct 2024 – Present

Solar Square Energy | Mumbai, India

- **Leading the ERP POD at SolarSquare**, architecting and operating a horizontally scaled, distributed, cloud-native ERP platform on AWS — Auto Scaling Group web tier (Gunicorn/Nginx/SocketIO), dedicated worker fleet, ElastiCache Redis, RDS MariaDB, S3, and Lambda — delivering enterprise-grade performance and elasticity for 450+ users across supply chain, finance, and warehouse operations (B2B & B2C).
- **Engineered and integrated mission-critical backend systems** — banking and payment gateway APIs, AMC lifecycle automation, and an Order Management System (OMS) — enabling end-to-end financial transactions and order automation across B2B and B2C channels.
- **Inventory tracking & loss prevention** — built workflows and data pipelines that identified and prevented ~₹20 Lakhs of mid-mile logistics losses, improving operational transparency and warehouse efficiency.
- **Implemented DevOps and observability best practices** by integrating Kibana for centralized logging and building an automated CI/CD pipeline with the GitHub API and LLM-powered (Claude, OpenCode) code reviews, improving developer velocity, code quality, and release efficiency.
- **Strengthened IT General Controls (ITGC) posture for IPO readiness** by building a custom Role-Based Access Control (RBAC) framework with end-to-end activity logging, implementing maker-checker approval workflows on sensitive financial transactions (enforcing segregation of duties), and developing an audit framework that captures access, change, and approval evidence into structured reports — directly addressing the access, change, and operations control domains expected in SOX and ITGC audits.

Associate Consultant

Jun 2022 – Oct 2024

DataCurate Technologies | Mumbai, India

Client: HDFC ERGO

- Engineered a supervised machine learning model to target WhatsApp campaigns at users with elevated lead-conversion probability, cutting messaging costs by 70% and significantly improving lead conversion efficacy.
- Collaborated with Marketing and Engineering teams to build a unified marketing data pipeline optimising email, SMS, and WhatsApp channels, contributing to a 6% lift in overall lead conversion.

Client: Star Health Insurance

- Built ETL workflows, a Detail Data Store (DDS), automation scripts, production reports, and analytical dashboards using SQL and SAS-based technologies in a production environment.
- Contributed to fraud-detection mechanisms leveraging data warehousing, rule engines, and custom rule sets exposed via APIs.

OPEN SOURCE CONTRIBUTIONS

frappe/erpnext — Core Stock Module Performance Optimization · [PR #51644](#)

World's leading open-source ERP platform · 33.9K+ GitHub stars · 11.3K+ forks

- Refactored `get_stock_ledgers_for_serial_nos` to leverage ERPNext v15's Serial and Batch Bundle data model, replacing application-layer filtering with bundle-based SQL joins for serial-number validation while preserving backward compatibility with legacy SLE serial data.
- Reduced query execution time from ~583s to ~67s (~8.7× faster) and database queries from ~104K to ~14.5K (~86% reduction) on serial-tracked datasets — directly improving Delivery Note, Purchase Receipt, Stock Entry, and Stock Reconciliation workflows for the global ERPNext community.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS

Frameworks & Platforms: Frappe, ERPNext, Flask, FastAPI

Cloud & Infrastructure: AWS (EC2, Auto Scaling Groups, RDS MariaDB, ElastiCache Redis, S3, Lambda, CloudWatch, IAM), Nginx, Linux / Bash, Docker

Data & Analytics: ETL, Data Pipelines, Analytics, SAS

DevOps & Tooling: Git / GitHub, CI/CD, Kibana, Supervisor

Other: REST APIs, System Design, GenAI / LLM Integration, DSA

PERSONAL PROJECTS

Real-Time Product Recommendation System for ERPNext POS

- Built a real-time product recommendation engine for ERPNext POS using the Apriori algorithm with Pandas and mlxtend, enabling intelligent upsell suggestions at the point of sale.

Amba Tech (NGO) — Event Management ERP

- Designed and developed an end-to-end ERP system with an Angular web app, mobile app, and Metabase dashboards for event management. Implemented a QR-based attendance system to enable precise tracking and data-driven decisions for the organising team.

CERTIFICATIONS & ACHIEVEMENTS

- **Microsoft Certified: Azure Fundamentals (AZ-900)** — Mar 2023
- **Winner — Mumbai Hackathon 2022**, hosted by Zerodha and Frappe — Mar 2022

EDUCATION

Bachelor of Engineering, Information Technology

Jul 2018 – Jun 2022

University of Mumbai | Mumbai, India